

“ We’ve been through four OS generations and multiple bus generations, but we’re still on the first version of BIOS. It’s become, increasingly, a barrier to innovation in the industry. ”

— MIKE RICHMOND

Manager of platform software in Intel’s Software and Solutions Group

only has to touch the things it needs to boot. It also helps the overall reliability of the computer. In the PC world, we put up with that stuff, and we shouldn’t.”

Long ago, the BIOS was relatively straightforward in its design, but over the years, it’s morphed into a tangled mess, being changed and updated as and when new technologies came up. The BIOS hangover is causing more and more problems, feels Mark Doran, Intel’s principal engineer behind EFI.

There are no standards for how a BIOS is created or configured. An example of the chaos that exists is that it’s not uncommon for extension boards from different manufacturers to clash in their use of the BIOS. And if people do use the BIOS to change configurations or diagnose a problem, there’s often not much help: Even experienced support staff are often perplexed by the BIOS.

EFI promises to change all that. It’s a tiny OS in its own right. The first difference you’ll see is in the splash screen: Unlike the BIOS, EFI supports high-resolution displays. It can run with a proper graphical user interface, rather than the

text-only, DOS-like interface. And EFI also has its own networking, so it can be used for remote diagnostics.

There’s a significant change from the coding point of view as well. The BIOS is the last place on the PC where programmers have to code in assembly language. EFI is almost entirely written in C, and allows additions to be created using standard programming tools. Such additions can include much more detailed and useful diagnostics, self-configuration programs, and ways to sort out problems even if the OS has hung. For example, if the OS hangs, you can look at the state of the machine, change the configuration, load a different driver, and restart.

And naturally, there will be legacy support: EFI can behave exactly like a BIOS, if required.

Intel’s Framework

Intel recently introduced an innovative concept in BIOS: the Platform Innovation Framework for EFI. This is a modular, platform-independent architecture that performs boot and other BIOS functions. The Framework allows PC makers to write

pre-boot software modules, similar to Windows drivers. When used with the EFI, the Framework eliminates the need for a BIOS. Intel thinks of this as a key boot architecture that will be modular, flexible, and scalable enough to last for the next 20 years or so.

BIOS vs the Framework

Components have changed several times since the BIOS was first developed. Processing power has increased 600 times, and hundreds of new peripherals have been introduced. Over the years, the BIOS has grown to deal with new components, larger and more complex chipsets, add-on cards, and so on. As it grew, it became increasingly unwieldy. The humble BIOS simply cannot handle the demands of cutting-edge architectures.

Intel’s answer to BIOS limitations has been EFI and the Framework, a modular architecture based on low-level drivers. It sits beneath the EFI, and performs boot and other BIOS functions in a faster, modular, more flexible way. It’s stored in firmware, so it’s more secure from viruses and other types of attack than past BIOSes.

Stop, Thief!

On 27th May 2003, Phoenix Technologies and Softex unveiled anti-theft software that uses BIOS-based technology to disable stolen machines, wipe data and trace the thief remotely via the Internet.

TheftGuard is software developed by Softex, and certified by Phoenix to work with its FirstBIOS chips and Core Managed Environment (CME) software. It aims to give computer makers—and their PC buyers—a more secure choice of anti-theft technology.

Unlike other, though somewhat similar anti-theft solutions, TheftGuard resides both in Phoenix’s FirstBIOS and in the host-protected area (HPA) section of the hard drive. Removing and replacing the

hard drive, a tactic that can be used to circumvent some anti-theft concepts, doesn’t get the thief anywhere with TheftGuard in place.

Another trick that can eradicate anti-theft software—running FDISK to reformat the drive—is also foiled by TheftGuard’s place in the HPA section of the hard drive, which is immune to such reformatting tools.

After a TheftGuard-equipped machine is reported as stolen to a special Web site, the stolen PC can be detected when it next connects to the Internet. At that time, the computer can be disabled, all data on its hard drive wiped clean, and an IP trace put on the connection to determine the physical location of the system. And

because TheftGuard is integrated with the BIOS, it can cripple the computer by not allowing it to boot.

In addition, a simple re-flash of the BIOS would not thwart TheftGuard: One would have to be extremely tech savvy to remove TheftGuard. So much so that most people who would know how to do this work for Phoenix! The part of the BIOS where TheftGuard resides is not updatable through a flash upgrade.

There are, of course, the privacy concerns of companies and end-users over putting a tracer on their machines. TheftGuard, however, is optional, and is not included with every BIOS that Phoenix produces and ships.